

SURIYA PRASATH PARAMESWARAN

DATA SCIENTIST

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[LinkedIn](#) · [GitHub](#) · [Portfolio](#)

PROFILE

Data scientist building end-to-end ML systems with a focus on production reliability and business ROI, holding an MSc in Data Science and Analytics from the University of Leeds. Project and dissertation work spans fraud detection, customer retention, marketing optimisation, and cybersecurity risk modelling, with strong attention to calibrated models, capacity-constrained decision frameworks, and shipped services. Comfortable taking problems from framing through deployment via FastAPI, Docker, and MLflow.

EXPERIENCE

Data Science Researcher · University of Leeds (MSc Dissertation)

Sep 2024 — Sep 2025

- Built an end-to-end NLP pipeline predicting CVE severity (LOW/MEDIUM/HIGH/CRITICAL) on 29,000 records, hitting F1 of 0.81 on the CRITICAL class, the highest-stakes tier
- Compared a fine-tuned SecBERT transformer against a TF-IDF + XGBoost baseline with SMOTE oversampling and SHAP token-level explainability; benchmarked LLM cost-quality tradeoffs
- Shipped as a Streamlit app with MLflow experiment tracking and GitHub Actions CI/CD

Tools Python · PyTorch · HuggingFace · SecBERT · XGBoost · SHAP · MLflow · Streamlit

Freelance Data Analyst · Self-Employed

2024 — Present

- Built a Power BI inventory and sales dashboard for a family-owned grocery shop with 200+ products; owner logs daily stock and sales, dashboard returns live profit, best-sellers, and a year-end performance summary
- Worked directly with a non-technical owner to translate reporting needs into a simple data-entry workflow with decision-ready output

IoT & Data Analytics Intern · Shree Kay Vee Automation

Jan 2023 — Apr 2023

- Analysed IoT sensor data from manufacturing equipment to identify patterns linked to machine performance and failure risk; built dashboards used by the operations team
- Improved data quality and reporting reliability through structured preprocessing and validation workflows

PROJECTS

Fraud Triage Engine

2025

- Reframed fraud detection as capacity-constrained ranking on 284,000 transactions; achieved 81.3% fraud recall at 500 daily reviews with a 70% drop in review cost
- Built a calibrated LightGBM + XGBoost ensemble with temporal splitting, cost-sensitive thresholding, and SHAP explainability; drift monitor triggers automated retraining alerts
- Shipped as a FastAPI service with Docker and pytest coverage for production reliability

Tools Python · LightGBM · XGBoost · SHAP · Isotonic Calibration · FastAPI · Docker

Telecom Churn Decision Engine

2025

- Built a profit-optimised retention engine combining T-learner uplift modelling, survival-adjusted CLV, and 0/1 knapsack MILP allocation; beat highest-churn-first targeting by 37.7% mean profit
- Modelled 8.15x ROI under a £10K budget with correlated Monte Carlo simulation across 2,000 scenarios, beating greedy targeting by £13,823 in the same simulation
- Shipped as a FastAPI service with Docker and Monte Carlo stress-testing baked in

Tools Python · LightGBM · scikit-learn · PuLP (CBC MILP) · Monte Carlo · FastAPI · Docker

EDUCATION

MSc Data Science and Analytics · University of Leeds

Sep 2024 — Sep 2025

Graduate research in machine learning, NLP, and risk modelling; strong academic performance in Machine Learning, Statistical Learning, and Natural Language Processing.

BE Mechanical Engineering · Easwari Engineering College

Jun 2018 — Mar 2022

SKILLS

PROGRAMMING & DATA

Python, SQL, pandas, Polars, NumPy, DuckDB, PySpark, PostgreSQL, MySQL, dbt

MACHINE LEARNING

scikit-learn, LightGBM, XGBoost, CatBoost, SHAP, Optuna, SMOTE, Uplift Modelling, Isotonic Calibration, A/B Testing

DEEP LEARNING & NLP

PyTorch, HuggingFace Transformers, SecBERT, TF-IDF, Transfer Learning

TIME SERIES & OPTIMISATION

Prophet, ARIMA, SARIMA, MILP (PuLP), Monte Carlo Simulation, Drift Monitoring, Temporal Validation

MLOPS & CLOUD

FastAPI, Docker, MLflow, GitHub Actions CI/CD, pytest, Streamlit, Apache Airflow, AWS, Azure, GCP